Experiment Report 1: Which Networks Carry My Packets?

CSC6410 | macOS, home Wi-Fi | 1 November 2026 | all times EST

I traced five websites twice from my Mac, one hour apart, mapped every public router address to the AS that announces it with the RIPEstat Data API, and compared my result with the AS paths that BGP route collectors see. Four of the five sites were served by a hosting or CDN company (Amazon, Cloudflare, Automattic, Fastly); only Wikipedia ended in its own network.

1 Plan and predictions

Saved at 04:41:06 EST on 1 November 2026, before any traceroute or RIPEstat query. I wrote these in a Claude Code session where I had already pasted the assignment, so they are AI-assisted guesses and I do not count them as unaided predictions. For the fifth site I chose www.python.org.

Target	IP hops	ASes	Networks I expected
www.cuw.edu	12	3	home ISP -> transit (Lumen/Cogent) -> AS53355 (CUW)
www.ox.ac.uk	18	4	ISP -> transit -> JANET (AS786) -> Oxford (AS3462 area)
www.wikipedia.org	10	2	ISP -> Wikimedia (AS14907 / AS43821)
www.nasa.gov	10	3	ISP -> Akamai (AS16625/AS20940) or Fastly; NASA site is on a CDN
www.python.org	10	2	ISP -> Fastly (AS54113)

2 Consultation and critique

I asked Claude how to map a hop address to an AS. It said to use RIPEstat network-info for the covering prefix and origin ASN, as-overview for the AS name, and looking-glass for collector paths, and to skip private and shared address space. I tested every one of these calls myself (the cached answers are in the script folder) and they behave as described in the assignment's API table.

What I thought was wrong or missing: (1) The first plan assumed the CUW website would end in CUW's own network. The data show it ends in Amazon CloudFront (AS16509), so the assignment's hint that CUW is AS53355 describes the university, not its web server. (2) The first traceroute settings (default UDP probes) stalled, see problem 1. (3) The first script version assumed every hop has an origin AS; that is false for some interface addresses (problem 3). (4) My prediction of a CUW-owned AS and a JANET path for Oxford were both wrong, because both sites sit behind commercial CDNs.

3 Execution and iterations

Environment. macOS, home Wi-Fi. Command: `traceroute -n -I -q 2 -w 2 -m 25 <target>`, all five targets in parallel. Run 1 started at 05:11 EST and run 2 at 06:16 EST, more than an hour later. My script `hop2as.py` (a separate .py file) reads the traceroute text, skips 10/8, 172.16/12, 192.168/16 and 100.64/10, calls network-info once per public hop, adds as-overview names, caches every answer in a file and collapses consecutive duplicate ASNs. With `--collectors` it also prints three looking-glass paths per target. The code is commented in my own words.

Commands I ran

Command (macOS, home Wi-Fi)	What it shows
traceroute -n -I -q 2 -w 2 -m 25 www.cuw.edu (and the other four targets, twice)	The routers on the way to each site and which ones answer.
python3 hop2as.py raw/mi1_www.cuw.edu.txt --redact	The AS path for one trace, with private hops skipped.
python3 hop2as.py --collectors 54.240.162.100 ...	The prefix, the origin AS and three BGP collector paths for each destination.
export REQUESTS_CA_BUNDLE=ca.pem	Lets Python trust the system certificates so RIPEstat answers (problem 2).

Terminal - macOS - home Wi-Fi

```
% date
Sun Nov 1 05:11:13 EST 2026
% traceroute -n -I -q 2 -w 2 -m 25 www.cuw.edu
traceroute: Warning: www.cuw.edu has multiple addresses; using 54.240.162.100
traceroute to d27hkn76rnzir7.cloudfront.net (54.240.162.100), 25 hops max, 48 byte packets
 1  * 192.168.x.x  4.913 ms
 2  10.10.16.1  12.057 ms  9.281 ms
 3  172.16.5.8  8.373 ms  10.984 ms
 4  192.168.247.202  22.600 ms
    192.168.247.200  7.801 ms
 5  192.168.230.213  7.450 ms  5.042 ms
 6  192.168.230.194  9.857 ms  24.261 ms
 7  192.168.59.112  8.008 ms
    192.168.59.110  8.210 ms
 8  * *
 9  * *
10  * *
11  * *
12  * *
13  54.240.162.100  13.336 ms  9.924 ms
```

Figure 1. traceroute (ICMP probes) from my Mac to www.cuw.edu on home Wi-Fi, Sunday 1 Nov 2026, EST.

Terminal - macOS

```
% python3 hop2as.py raw/mi1_www.cuw.edu.txt --redact
hop  ip              prefix              AS        holder
 13  54.240.162.100   54.240.162.0/24     AS16509   AMAZON-02 - Amazon.com, Inc.
AS path: AS16509
Names  : AMAZON-02 - Amazon.com, Inc.
IP hops answered up to: 13 (+0 silent trailing)   AS hops: 1

% python3 hop2as.py raw/mi1_www.wikipedia.org.txt --redact
hop  ip              prefix              AS        holder
 10  103.198.x.x      103.198.x.x/24     ISP-A-intl   my ISP, international network
 11  103.198.x.x   103.198.x.x/24     ISP-A-intl   my ISP, international network
 12  103.198.x.x   103.198.x.x/24     ISP-A-intl   my ISP, international network
 13  27.111.x.x   -                    ?         no origin AS; whois: colocation / exchange network
 14  103.102.x.x  103.102.x.x/24     AS14907   WIKIMEDIA - Wikimedia Foundation Inc.
 15  103.102.x.x  103.102.x.x/24     AS14907   WIKIMEDIA - Wikimedia Foundation Inc.
AS path: ISP-A-intl -> AS14907
Names  : my ISP, international network -> WIKIMEDIA - Wikimedia Foundation Inc.
IP hops answered up to: 15 (+0 silent trailing)   AS hops: 2
```

Figure 2. My script running on two traces. It skips private hops and prints the AS path (here AS16509 for CUW, and ISP-A-intl then AS14907 for Wikipedia).

RIPEstat network-info - 103.102.x.x (Wikipedia destination)

```
{
  "asns": [
    "14907"
  ],
  "prefix": "103.102.x.x/24"
}
```

Figure 3. One RIPEstat answer: the covering prefix and origin AS for the Wikipedia destination address (script printout of the API's data field).

Problems and fixes (symptom, change, evidence)

#	Symptom	What I changed	Evidence
1	The default UDP probes went silent: CUW stopped after hop 7, Wikipedia after hop 12, python.org after hop 10.	Added -I so traceroute sends ICMP echo probes.	Fig 4: all five targets then answered from the destination itself (hops 11 to 15).
2	Every RIPEstat call failed with CERTIFICATE_VERIFY_FAILED (self-signed certificate in chain).	Exported the macOS system certificates to a file and pointed Python at it with REQUESTS_CA_BUNDLE.	Fig 5: the same command then printed the prefixes and AS numbers.
3	Wikipedia hop 13 had no origin AS: network-info returned an empty list.	Added a whois fallback that prints the registered block name when BGP has no origin.	Fig 6: the hop now reads as a colocation/exchange block instead of a blank.

Before: UDP probes (default)

```
% traceroute -n -q 2 -w 2 -m 25 www.cuw.edu
traceroute: Warning: www.cuw.edu has multiple
addresses; using 54.240.162.100
traceroute to d27hkn76rnzir7.cloudfront.net
(54.240.162.100), 25 hops max, 40 byte packets
 1  * 192.168.x.x  3.033 ms
 2  10.10.16.1  6.832 ms  10.235 ms
 3  172.16.5.8  8.728 ms
    172.16.5.10  8.363 ms
 4  192.168.247.204  8.148 ms
    192.168.247.206  7.467 ms
 5  192.168.230.213  10.122 ms  14.331 ms
 6  192.168.230.195  9.267 ms  8.932 ms
 7  192.168.59.116  6.636 ms
    192.168.59.110  39.898 ms
  ... (hops 8-25: * *)
```

After: ICMP probes (-I)

```
% traceroute -n -I -q 2 -w 2 -m 25 www.cuw.edu
traceroute: Warning: www.cuw.edu has multiple
addresses; using 54.240.162.100
traceroute to d27hkn76rnzir7.cloudfront.net
(54.240.162.100), 25 hops max, 48 byte packets
 1  * 192.168.x.x  4.913 ms
 2  10.10.16.1  12.057 ms  9.281 ms
 3  172.16.5.8  8.373 ms  10.984 ms
 4  192.168.247.202  22.600 ms
    192.168.247.200  7.801 ms
 5  192.168.230.213  7.450 ms  5.042 ms
 6  192.168.230.194  9.857 ms  24.261 ms
 7  192.168.59.112  8.008 ms
    192.168.59.110  8.210 ms
 8  * *
 9  * *
10  * *
11  * *
12  * *
13  54.240.162.100  13.336 ms  9.924 ms
```

Figure 4. Problem 1. Before: default UDP probes stopped answering after hop 7 for CUW. After: with -I the same target answered at hop 13.

Before

```
% python3 hop2as.py raw/m1_www.wikipedia.org.txt
[warn] network-info 103.198.x.x:
HTTPSConnectionPool(host='stat.ripe.net', port=443):
Max retries exceeded
with url: /data/network-info/data.json?
resource=103.198.x.x (Caused by
SSLError(SSLCertVerificationError(1,
'[SSL: CERTIFICATE_VERIFY_FAILED] certificate verify
failed: self-signed certificate in certificate
chain')))
lookup failed for every public hop
```

After

```
% export REQUESTS_CA_BUNDLE=$PWD/ca.pem
% python3 hop2as.py raw/mi1_www.wikipedia.org.txt
hop  ip              prefix              AS
holder
 10  103.198.x.x      103.198.x.x/24     ISP-A-intl
my ISP, international network
 11  103.198.x.x   103.198.x.x/24     ISP-A-intl
my ISP, international network
 12  103.198.x.x   103.198.x.x/24     ISP-A-intl
my ISP, international network
 13  27.111.x.x   -                    ?         no
origin AS; whois: colocation / exchange network
```

Figure 5. Problem 2. Before: every RIPEstat call failed certificate verification. After: pointing Python at the system certificates made the lookups work.

Before: RIPEstat answer for the hop

```
{"asns": [], "prefix": ""}

hop 13  27.111.x.x   ?   no origin AS
```

After: whois fallback

```
 13  27.111.x.x   -                    ?         no
origin AS; whois: colocation / exchange network
```

Figure 6. Problem 3. Before: hop 13 on the way to Wikipedia had no origin AS in BGP. After: the whois fallback names the block (a colocation/exchange network), so the gap is explained.

4 Results from my own data

ISP-A, ISP-A-intl and ISP-B are my own internet provider's networks. "IP hops" is the last hop number that answered; the number in brackets is how many hop numbers returned at least one reply.

www.cuw.edu

Run	Start	Destination	IP hops (replies)	AS path from traceroute	Origin AS (RIPEstat)	Last AS = origin?	Traceroute ASes seen in collector paths
1	05:11 EST	54.240.162.100	13 (8 answered)	AS16509	AS16509	yes	AS16509
2	06:16 EST	54.240.162.50	17 (9 answered)	AS16509	AS16509	yes	AS16509

Route collector (prefix 54.240.162.0/24)	AS path seen by the collector
RRC01 London, GB	15692 2914 16509 16509 16509
RRC00 Amsterdam, NL	59919 9002 6453 16509 16509 16509
RRC24 Montevideo, UY	46997 38008 38008 174 16509 16509 16509

www.ox.ac.uk

Run	Start	Destination	IP hops (replies)	AS path from traceroute	Origin AS (RIPEstat)	Last AS = origin?	Traceroute ASes seen in collector paths
1	05:11 EST	172.66.169.161	11 (9 answered)	ISP-A -> AS13335	AS13335	yes	AS13335
2	06:16 EST	172.66.169.161	11 (9 answered)	ISP-A -> AS13335	AS13335	yes	AS13335

Route collector (prefix 172.66.160.0/20)	AS path seen by the collector
RRC01 London, GB	8218 13335
RRC00 Amsterdam, NL	59919 13335
RRC24 Montevideo, UY	46997 38008 38008 13335

www.wikipedia.org

Run	Start	Destination	IP hops (replies)	AS path from traceroute	Origin AS (RIPEstat)	Last AS = origin?	Traceroute ASes seen in collector paths
1	05:11 EST	103.102.x.x	15 (13 answered)	ISP-A-intl -> AS14907	AS14907	yes	AS14907
2	06:16 EST	103.102.x.x	15 (13 answered)	ISP-A-intl -> AS14907	AS14907	yes	AS14907

Route collector (prefix 103.102.x.x/24)	AS path seen by the collector
RRC01 London, GB	8218 6461 6453 14907
RRC00 Amsterdam, NL	59919 9002 6453 14907
RRC24 Montevideo, UY	46997 38008 38008 3491 6453 14907

www.nasa.gov

Run	Start	Destination	IP hops (replies)	AS path from traceroute	Origin AS (RIPEstat)	Last AS = origin?	Traceroute ASes seen in collector paths
1	05:11 EST	192.0.66.108	13 (10 answered)	ISP-B -> AS2635	AS2635	yes	AS2635
2	06:16 EST	192.0.66.108	13 (10 answered)	ISP-B -> AS2635	AS2635	yes	AS2635

Route collector (prefix 192.0.66.0/24)	AS path seen by the collector
RRC01 London, GB	15692 2914 2635
RRC00 Amsterdam, NL	59919 2635
RRC24 Montevideo, UY	46997 38008 38008 2635

www.python.org

Run	Start	Destination	IP hops (replies)	AS path from traceroute	Origin AS (RIPEstat)	Last AS = origin?	Traceroute ASes seen in collector paths
1	05:11 EST	151.101.128.223	11 (9 answered)	ISP-A -> AS54113	AS54113	yes	AS54113
2	06:16 EST	151.101.64.223	11 (9 answered)	ISP-A -> AS54113	AS54113	yes	AS54113

Route collector (prefix 151.101.128.0/22)	AS path seen by the collector
RRC01 London, GB	8218 54113
RRC00 Amsterdam, NL	59919 54113
RRC24 Montevideo, UY	46997 38008 38008 3491 54113

5 Analysis

Target	Predicted IP / AS	Run 1 IP / AS	Run 2 IP / AS
www.cuw.edu	12 / 3	13 / 1	17 / 1
www.ox.ac.uk	18 / 4	11 / 2	11 / 2
www.wikipedia.org	10 / 2	15 / 2	15 / 2
www.nasa.gov	10 / 3	13 / 2	13 / 2
www.python.org	10 / 2	11 / 2	11 / 2

Predictions vs results. My hop counts were off for every site. I guessed 10 to 12 hops for four sites and got 11 to 17; I guessed 18 for Oxford and got 11. My AS guesses were too high for CUW (3 guessed, 1 seen), Oxford (4 guessed, 2 seen) and NASA (3 guessed, 2 seen). The Wikipedia and python.org guesses (2 ASes each, ending in Wikimedia and Fastly) were right. I expected CUW's own network, JANET for Oxford and Akamai for NASA. The real ends were Amazon CloudFront (AS16509), Cloudflare (AS13335) and Automattic (AS2635). The websites of a university and a government agency both sit in hosting companies, so the institution's name tells me little about the AS at the end.

IP hops vs AS hops. I counted 11 to 17 IP hops but only 1 or 2 ASes for every site. The reason is that hops 1 to 7 were all private addresses (my router and my provider's inside network), which are not in BGP and cannot be mapped. For CUW the next five hops (8 to 12) did not answer at all, so the only AS I could see was the destination's. For the others, my provider's last visible hop (hop 10 for Oxford and python.org, hop 12 for NASA) was followed directly by the destination network: Cloudflare for Oxford, Fastly for python.org and Automattic for NASA. For Wikipedia I saw two hand-offs: my provider's international network (hops 10 to 12), a colocation block with no AS in BGP (hop 13, found with whois), then Wikimedia (AS14907, hops 14 and 15). Run 2, an hour later, gave the same AS paths for all five sites. Only the CDN address changed (CUW answered from 54.240.162.50

and needed 17 hops instead of 13), which shows the number of IP hops depends on which server the name points to.

Traceroute vs BGP. In all ten runs the last AS in my traceroute was the same as the origin AS that the three route collectors report. My provider's ASes never appear in the collector paths, because the collectors are in London, Amsterdam and Montevideo and look at the prefix from there. So the destination matches and the middle does not. That is expected: traceroute shows only the forward path from my Mac, and only routers that reply.

Where I see routing policy. Amazon's AS16509 is repeated three times in all three CUW collector paths (for example 59919 9002 6453 16509 16509 16509), and AS38008 appears twice in the Montevideo path. This looks like AS-path prepending: an AS adds itself several times so the path looks longer and other routes are preferred. I cannot tell the reason from my data. The paths to the same prefix are also different lengths. For Wikipedia the London path has 4 ASes and the Montevideo path has 6 entries (5 different ASes), while my provider reached it in 2. Each collector's network picks by its own policy, not by the shortest path.

Link to the lecture. Each destination sat in one announced prefix (a /24, /22 or /20), and a router sends the packet using the longest matching prefix. Between networks BGP picks by policy, such as local preference and path length, which explains the prepending and the different paths. Inside my provider's network the private hops are where OSPF would choose the least-cost path; I cannot see those costs from outside.

6 Pitfalls and limits

A traceroute shows only the forward path from my Mac, and only routers that answer; many replies were private addresses or timeouts, so AS counts are lower bounds. The AS of a router interface is the AS that announces that address, which for a link between two networks can be the neighbour's AS. The collectors look at the prefix from London, Amsterdam and Montevideo, not from my house, so their paths cannot match mine exactly. CDN names return different addresses on different runs. If I did it again I would capture the traceroute and the API answer in one sitting, and I would try TCP probes as a third method.

7 Lessons and reflection on the help I received

The AI saved time on the script structure and on mapping addresses with RIPEstat. It was wrong or too optimistic about the traceroute settings, about every hop having an AS, and about the institution's own AS being the end of the path. I only found these by running the commands and reading the output, which is why each one is in section 3 with before and after evidence.

My script is submitted separately as hop2as.py.

Appendix 1 Sources consulted

Course assignment sheet (API table for network-info, as-overview, looking-glass, bgp-state).
RIPEstat Data API, endpoints called: https://stat.ripe.net/data/network-info/data.json, /as-overview/, /looking-glass/, /whois/ (checked by calling them and reading the JSON).
AI tool: Claude (Claude Code session on my Mac), documented in AI Report 1.

Appendix 2 Raw traceroute and script output

As printed; times are EST.

```
% date -u  (run 1: 2026-11-01 05:11:13 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.cuw.edu
traceroute: Warning: www.cuw.edu has multiple addresses; using 54.240.162.100
traceroute to d27hkn76rnzir7.cloudfront.net (54.240.162.100), 25 hops max, 48 byte packets
 1  * 192.168.x.x  4.913 ms
 2  10.10.16.1  12.057 ms  9.281 ms
 3  172.16.5.8  8.373 ms  10.984 ms
 4  192.168.247.202  22.600 ms
    192.168.247.200  7.801 ms
 5  192.168.230.213  7.450 ms  5.042 ms
 6  192.168.230.194  9.857 ms  24.261 ms
 7  192.168.59.112  8.008 ms
    192.168.59.110  8.210 ms
 8  * *
 9  * *
10  * *
11  * *
12  * *
13  54.240.162.100  13.336 ms  9.924 ms
```

```
% date -u  (run 1: 2026-11-01 05:11:13 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.ox.ac.uk
traceroute: Warning: www.ox.ac.uk has multiple addresses; using 172.66.169.161
traceroute to www.ox.ac.uk.cdn.cloudflare.net (172.66.169.161), 25 hops max, 48 byte packets
 1  192.168.x.x  549.993 ms  8.101 ms
 2  10.10.16.1  11.545 ms  6.179 ms
 3  172.16.5.8  25.197 ms  37.112 ms
 4  192.168.247.200  13.083 ms  9.585 ms
 5  192.168.230.213  13.874 ms  16.121 ms
 6  192.168.230.195  25.734 ms  6.105 ms
 7  192.168.59.114  7.327 ms  21.741 ms
 8  * *
 9  * *
10  49.44.x.x  62.564 ms  17.324 ms
11  172.66.169.161  17.357 ms  16.615 ms
```

```
% date -u  (run 1: 2026-11-01 05:11:13 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.wikipedia.org
traceroute to dyna.wikimedia.org (103.102.x.x), 25 hops max, 48 byte packets
 1  192.168.x.x  543.151 ms  8.130 ms
 2  10.10.16.1  12.458 ms  26.402 ms
 3  172.16.5.8  41.166 ms  6.245 ms
 4  192.168.247.200  6.343 ms
    192.168.247.202  10.748 ms
 5  192.168.230.213  12.762 ms  12.919 ms
 6  192.168.230.195  22.327 ms  7.028 ms
 7  192.168.59.116  8.617 ms  6.976 ms
 8  * *
 9  * *
10  103.198.x.x  23.251 ms  47.913 ms
11  103.198.x.x  167.443 ms  132.953 ms
12  103.198.x.x  59.948 ms  48.811 ms
13  27.111.x.x  60.948 ms  51.128 ms
14  103.102.x.x  51.781 ms  49.570 ms
15  103.102.x.x  56.023 ms  99.163 ms
```

```
% date -u  (run 1: 2026-11-01 05:11:13 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.nasa.gov
traceroute to nasa-gov.go-vip.net (192.0.66.108), 25 hops max, 48 byte packets
 1  192.168.x.x  5.618 ms  3.262 ms
 2  10.10.16.1  7.565 ms  26.418 ms
```

```
 3  172.16.5.10  43.730 ms  8.771 ms
 4  192.168.247.200  7.042 ms  9.109 ms
 5  192.168.230.213  21.537 ms  20.281 ms
 6  192.168.230.195  9.101 ms  8.469 ms
 7  192.168.59.112  8.613 ms
    192.168.59.110  19.605 ms
 8  * *
 9  * *
10  202.191.x.x  21.793 ms  20.726 ms
11  * *
12  202.191.x.x  30.096 ms  18.388 ms
13  192.0.66.108  17.466 ms  18.719 ms
```

```
% date -u  (run 1: 2026-11-01 05:11:13 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.python.org
traceroute: Warning: www.python.org has multiple addresses; using 151.101.128.223
traceroute to dualstack.python.map.fastly.net (151.101.128.223), 25 hops max, 48 byte packets
 1  * 192.168.x.x  7.880 ms
 2  10.10.16.1  5.662 ms  6.752 ms
 3  172.16.5.10  8.581 ms  9.474 ms
 4  192.168.247.206  12.009 ms  12.001 ms
 5  192.168.230.213  7.202 ms  6.527 ms
 6  192.168.230.195  12.063 ms  15.709 ms
 7  192.168.59.114  9.121 ms
    192.168.59.116  6.815 ms
 8  * *
 9  * *
10  49.44.x.x  10.123 ms  52.718 ms
11  151.101.128.223  16.144 ms  9.046 ms
```

```
% date -u  (run 2: 2026-11-01 06:16:37 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.cuw.edu
traceroute: Warning: www.cuw.edu has multiple addresses; using 54.240.162.50
traceroute to d27hkn76rnzir7.cloudfront.net (54.240.162.50), 25 hops max, 48 byte packets
 1  * 192.168.x.x  3.874 ms
 2  10.10.16.1  6.475 ms  7.255 ms
 3  172.16.5.10  8.960 ms  5.811 ms
 4  192.168.247.200  6.682 ms
    192.168.247.202  9.225 ms
 5  192.168.230.213  6.157 ms  6.339 ms
 6  192.168.230.194  6.488 ms  6.330 ms
 7  192.168.59.114  6.872 ms  6.086 ms
 8  * *
 9  * *
10  * *
11  49.44.x.x  24.452 ms  18.448 ms
12  * *
13  * *
14  * *
15  * *
16  * *
17  54.240.162.50  41.757 ms  18.563 ms
```

```
% date -u  (run 2: 2026-11-01 06:16:37 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.ox.ac.uk
traceroute: Warning: www.ox.ac.uk has multiple addresses; using 172.66.169.161
traceroute to www.ox.ac.uk.cdn.cloudflare.net (172.66.169.161), 25 hops max, 48 byte packets
 1  * 192.168.x.x  3.785 ms
 2  10.10.16.1  8.376 ms  6.118 ms
 3  172.16.5.8  7.675 ms  7.429 ms
 4  192.168.247.200  10.765 ms  9.794 ms
 5  192.168.230.213  10.571 ms  8.419 ms
 6  192.168.230.195  6.662 ms  7.315 ms
 7  192.168.59.116  9.210 ms  5.499 ms
 8  * *
 9  * *
10  49.44.x.x  67.546 ms  24.736 ms
11  172.66.169.161  17.668 ms  17.393 ms
```

```
% date -u  (run 2: 2026-11-01 06:16:37 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.wikipedia.org
traceroute to dyna.wikimedia.org (103.102.x.x), 25 hops max, 48 byte packets
 1  192.168.x.x  30.819 ms  8.077 ms
 2  10.10.16.1  15.774 ms  7.007 ms
 3  172.16.5.8  18.135 ms  8.840 ms
 4  192.168.247.202  8.383 ms  6.840 ms
 5  192.168.230.213  9.051 ms  10.343 ms
 6  192.168.230.194  24.466 ms  7.133 ms
```

```
 7  192.168.59.114  10.378 ms
    192.168.59.116  9.680 ms
 8  * *
 9  * *
10  103.198.x.x  54.997 ms  19.117 ms
11  103.198.x.x  57.873 ms  51.775 ms
12  103.198.x.x  53.720 ms  48.376 ms
13  27.111.x.x  57.135 ms  49.629 ms
14  103.102.x.x  54.428 ms  48.984 ms
15  103.102.x.x  55.963 ms  53.462 ms
```

```
% date -u  (run 2: 2026-11-01 06:16:37 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.nasa.gov
traceroute to nasa-gov.go-vip.net (192.0.66.108), 25 hops max, 48 byte packets
 1  192.168.x.x  8.826 ms  3.488 ms
 2  10.10.16.1  14.861 ms  7.730 ms
 3  172.16.5.10  7.722 ms  9.029 ms
 4  192.168.247.202  6.225 ms  6.563 ms
 5  192.168.230.213  27.714 ms  5.604 ms
 6  192.168.230.195  11.385 ms  9.977 ms
 7  192.168.59.112  9.424 ms  6.983 ms
 8  * *
 9  * *
10  202.191.x.x  48.746 ms  20.241 ms
11  * *
12  202.191.x.x  22.404 ms  20.755 ms
13  192.0.66.108  19.389 ms  19.753 ms
```

```
% date -u  (run 2: 2026-11-01 06:16:37 EST)
% traceroute -n -I -q 2 -w 2 -m 25 www.python.org
traceroute: Warning: www.python.org has multiple addresses; using 151.101.64.223
traceroute to dualstack.python.map.fastly.net (151.101.64.223), 25 hops max, 48 byte packets
 1  192.168.x.x  40.282 ms  9.757 ms
 2  10.10.16.1  6.429 ms  9.383 ms
 3  172.16.5.10  18.791 ms  10.286 ms
 4  192.168.247.206  8.331 ms
    192.168.247.204  9.072 ms
 5  192.168.230.213  7.583 ms  27.714 ms
 6  192.168.230.194  9.546 ms  7.180 ms
 7  192.168.59.116  12.476 ms
    192.168.59.114  13.024 ms
 8  * *
 9  * *
10  49.44.x.x  30.930 ms  7.920 ms
11  151.101.64.223  31.182 ms  8.395 ms
```

Script output

Run 1

```
% python3 hop2as.py raw/mi1_www.cuw.edu.txt --redact
hop  ip              prefix              AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1      private/shared       -         skipped (not in BGP)
  3  172.16.5.8      private/shared       -         skipped (not in BGP)
  4  192.168.247.202  private/shared      -         skipped (not in BGP)
  5  192.168.230.213  private/shared      -         skipped (not in BGP)
  6  192.168.230.194  private/shared      -         skipped (not in BGP)
  7  192.168.59.112   private/shared      -         skipped (not in BGP)
 13  54.240.162.100   54.240.162.0/24     AS16509   AMAZON-02 - Amazon.com, Inc.

AS path: AS16509
Names  : AMAZON-02 - Amazon.com, Inc.
IP hops answered up to: 13 (+0 silent trailing)   AS hops: 1


% python3 hop2as.py raw/mi1_www.ox.ac.uk.txt --redact
hop  ip              prefix              AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1      private/shared       -         skipped (not in BGP)
  3  172.16.5.8      private/shared       -         skipped (not in BGP)
  4  192.168.247.200  private/shared      -         skipped (not in BGP)
  5  192.168.230.213  private/shared      -         skipped (not in BGP)
  6  192.168.230.195  private/shared      -         skipped (not in BGP)
  7  192.168.59.114   private/shared      -         skipped (not in BGP)
 10  49.44.x.x        49.44.x.x/17        ISP-A   my ISP
 11  172.66.169.161   172.66.160.0/20     AS13335   CLOUDFLARENET - Cloudflare, Inc.
```

```
AS path: ISP-A -> AS13335
Names  : my ISP -> CLOUDFLARENET - Cloudflare, Inc.
IP hops answered up to: 11 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py raw/mi1_www.wikipedia.org.txt --redact
hop  ip               prefix               AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.8       private/shared       -         skipped (not in BGP)
  4  192.168.247.200  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.195  private/shared       -         skipped (not in BGP)
  7  192.168.59.116   private/shared       -         skipped (not in BGP)
 10  103.198.x.x      103.198.x.x/24     ISP-A-intl   my ISP, international network
 11  103.198.x.x   103.198.x.x/24     ISP-A-intl   my ISP, international network
 12  103.198.x.x   103.198.x.x/24     ISP-A-intl   my ISP, international network
 13  27.111.x.x   -                    ?         no origin AS; whois: colocation / exchange network
 14  103.102.x.x  103.102.x.x/24     AS14907   WIKIMEDIA - Wikimedia Foundation Inc.
 15  103.102.x.x  103.102.x.x/24     AS14907   WIKIMEDIA - Wikimedia Foundation Inc.

AS path: ISP-A-intl -> AS14907
Names  : my ISP, international network -> WIKIMEDIA - Wikimedia Foundation Inc.
IP hops answered up to: 15 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py raw/mi1_www.nasa.gov.txt --redact
hop  ip               prefix               AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.10      private/shared       -         skipped (not in BGP)
  4  192.168.247.200  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.195  private/shared       -         skipped (not in BGP)
  7  192.168.59.112   private/shared       -         skipped (not in BGP)
 10  202.191.x.x      202.191.x.x/23     ISP-B    my ISP, upstream provider
 12  202.191.x.x   202.191.x.x/22     ISP-B    my ISP, upstream provider
 13  192.0.66.108     192.0.66.0/24        AS2635    AUTOMATTIC - Automattic, Inc

AS path: ISP-B -> AS2635
Names  : my ISP, upstream provider -> AUTOMATTIC - Automattic, Inc
IP hops answered up to: 13 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py raw/mi1_www.python.org.txt --redact
hop  ip               prefix               AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.10      private/shared       -         skipped (not in BGP)
  4  192.168.247.206  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.195  private/shared       -         skipped (not in BGP)
  7  192.168.59.114   private/shared       -         skipped (not in BGP)
 10  49.44.x.x        49.44.x.x/22        ISP-A   my ISP
 11  151.101.128.223  151.101.128.0/22     AS54113   FASTLY - Fastly, Inc.

AS path: ISP-A -> AS54113
Names  : my ISP -> FASTLY - Fastly, Inc.
IP hops answered up to: 11 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py --collectors <destination addresses>

54.240.162.100  prefix 54.240.162.0/24  origin AS16509 (AMAZON-02 - Amazon.com, Inc.)
  RRC01 (London, GB): 15692 2914 16509 16509 16509
  RRC00 (Amsterdam, NL): 59919 9002 6453 16509 16509 16509
  RRC24 (Montevideo, UY): 46997 38008 38008 174 16509 16509 16509

172.66.169.161  prefix 172.66.160.0/20  origin AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
  RRC01 (London, GB): 8218 13335
  RRC00 (Amsterdam, NL): 59919 13335
  RRC24 (Montevideo, UY): 46997 38008 38008 13335

103.102.x.x  prefix 103.102.x.x/24  origin AS14907 (WIKIMEDIA - Wikimedia Foundation Inc.)
  RRC01 (London, GB): 8218 6461 6453 14907
  RRC00 (Amsterdam, NL): 59919 9002 6453 14907
  RRC24 (Montevideo, UY): 46997 38008 38008 3491 6453 14907

192.0.66.108  prefix 192.0.66.0/24  origin AS2635 (AUTOMATTIC - Automattic, Inc)
  RRC01 (London, GB): 15692 2914 2635
  RRC00 (Amsterdam, NL): 59919 2635
```

```
  RRC24 (Montevideo, UY): 46997 38008 38008 2635

151.101.128.223  prefix 151.101.128.0/22  origin AS54113 (FASTLY - Fastly, Inc.)
  RRC01 (London, GB): 8218 54113
  RRC00 (Amsterdam, NL): 59919 54113
  RRC24 (Montevideo, UY): 46997 38008 38008 3491 54113
```

Run 2

```
% python3 hop2as.py raw/mi2_www.cuw.edu.txt --redact
hop  ip               prefix              AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.10      private/shared       -         skipped (not in BGP)
  4  192.168.247.200  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.194  private/shared       -         skipped (not in BGP)
  7  192.168.59.114   private/shared       -         skipped (not in BGP)
 11  49.44.x.x        -                   ?         no origin AS; whois: RELIANCEJIO-IN
 17  54.240.162.50    54.240.162.0/24      AS16509   AMAZON-02 - Amazon.com, Inc.

AS path: AS16509
Names  : AMAZON-02 - Amazon.com, Inc.
IP hops answered up to: 17 (+0 silent trailing)   AS hops: 1


% python3 hop2as.py raw/mi2_www.ox.ac.uk.txt --redact
hop  ip               prefix              AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.8       private/shared       -         skipped (not in BGP)
  4  192.168.247.200  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.195  private/shared       -         skipped (not in BGP)
  7  192.168.59.116   private/shared       -         skipped (not in BGP)
 10  49.44.x.x        49.44.x.x/17       ISP-A   my ISP
 11  172.66.169.161   172.66.160.0/20      AS13335   CLOUDFLARENET - Cloudflare, Inc.

AS path: ISP-A -> AS13335
Names  : my ISP -> CLOUDFLARENET - Cloudflare, Inc.
IP hops answered up to: 11 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py raw/mi2_www.wikipedia.org.txt --redact
hop  ip               prefix              AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.8       private/shared       -         skipped (not in BGP)
  4  192.168.247.202  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.194  private/shared       -         skipped (not in BGP)
  7  192.168.59.114   private/shared       -         skipped (not in BGP)
 10  103.198.x.x      103.198.x.x/24     ISP-A-intl   my ISP, international network
 11  103.198.x.x   103.198.x.x/24     ISP-A-intl   my ISP, international network
 12  103.198.x.x   103.198.x.x/24     ISP-A-intl   my ISP, international network
 13  27.111.x.x   -                    ?         no origin AS; whois: colocation / exchange network
 14  103.102.x.x  103.102.x.x/24     AS14907   WIKIMEDIA - Wikimedia Foundation Inc.
 15  103.102.x.x  103.102.x.x/24     AS14907   WIKIMEDIA - Wikimedia Foundation Inc.

AS path: ISP-A-intl -> AS14907
Names  : my ISP, international network -> WIKIMEDIA - Wikimedia Foundation Inc.
IP hops answered up to: 15 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py raw/mi2_www.nasa.gov.txt --redact
hop  ip               prefix              AS        holder
  1  192.168.x.x     private/shared       -         skipped (not in BGP)
  2  10.10.16.1       private/shared       -         skipped (not in BGP)
  3  172.16.5.10      private/shared       -         skipped (not in BGP)
  4  192.168.247.202  private/shared       -         skipped (not in BGP)
  5  192.168.230.213  private/shared       -         skipped (not in BGP)
  6  192.168.230.195  private/shared       -         skipped (not in BGP)
  7  192.168.59.112   private/shared       -         skipped (not in BGP)
 10  202.191.x.x      202.191.x.x/23     ISP-B    my ISP, upstream provider
 12  202.191.x.x   202.191.x.x/22     ISP-B    my ISP, upstream provider
 13  192.0.66.108     192.0.66.0/24        AS2635    AUTOMATTIC - Automattic, Inc

AS path: ISP-B -> AS2635
Names  : my ISP, upstream provider -> AUTOMATTIC - Automattic, Inc
IP hops answered up to: 13 (+0 silent trailing)   AS hops: 2
```

```
% python3 hop2as.py raw/mi2_www.python.org.txt --redact
hop  ip              prefix              AS        holder
  1  192.168.x.x     private/shared      -         skipped (not in BGP)
  2  10.10.16.1       private/shared      -         skipped (not in BGP)
  3  172.16.5.10      private/shared      -         skipped (not in BGP)
  4  192.168.247.206  private/shared      -         skipped (not in BGP)
  5  192.168.230.213  private/shared      -         skipped (not in BGP)
  6  192.168.230.194  private/shared      -         skipped (not in BGP)
  7  192.168.59.116   private/shared      -         skipped (not in BGP)
 10  49.44.x.x        49.44.x.x/22        ISP-A   my ISP
 11  151.101.64.223   151.101.64.0/22      AS54113   FASTLY - Fastly, Inc.

AS path: ISP-A -> AS54113
Names  : my ISP -> FASTLY - Fastly, Inc.
IP hops answered up to: 11 (+0 silent trailing)   AS hops: 2


% python3 hop2as.py --collectors <destination addresses>

54.240.162.50  prefix 54.240.162.0/24  origin AS16509 (AMAZON-02 - Amazon.com, Inc.)
  RRC01 (London, GB): 15692 2914 16509 16509 16509
  RRC00 (Amsterdam, NL): 59919 9002 6453 16509 16509 16509
  RRC24 (Montevideo, UY): 46997 38008 38008 174 16509 16509 16509

172.66.169.161  prefix 172.66.160.0/20  origin AS13335 (CLOUDFLARENET - Cloudflare, Inc.)
  RRC01 (London, GB): 8218 13335
  RRC00 (Amsterdam, NL): 59919 13335
  RRC24 (Montevideo, UY): 46997 38008 38008 13335

103.102.x.x  prefix 103.102.x.x/24  origin AS14907 (WIKIMEDIA - Wikimedia Foundation Inc.)
  RRC01 (London, GB): 8218 6461 6453 14907
  RRC00 (Amsterdam, NL): 59919 9002 6453 14907
  RRC24 (Montevideo, UY): 46997 38008 38008 3491 6453 14907

192.0.66.108  prefix 192.0.66.0/24  origin AS2635 (AUTOMATTIC - Automattic, Inc)
  RRC01 (London, GB): 15692 2914 2635
  RRC00 (Amsterdam, NL): 59919 2635
  RRC24 (Montevideo, UY): 46997 38008 38008 2635

151.101.64.223  prefix 151.101.64.0/22  origin AS54113 (FASTLY - Fastly, Inc.)
  RRC01 (London, GB): 8218 54113
  RRC00 (Amsterdam, NL): 59919 54113
  RRC24 (Montevideo, UY): 46997 38008 38008 3491 54113
```

Appendix 3 Lab environment check

Terminal - lab check on macOS

```
% uname -sr
Darwin 25.6.0
% python3 --version
Python 3.14.5
% docker --version
Docker version 29.9.2, build a82e528
% docker run --rm hello-world
docker: Error response from daemon: Sign in to continue using Docker Desktop. [rest of message trimmed]
```

Figure 7. Lab check on my Mac: Python and the Docker CLI are installed, but Docker Desktop will not run containers until I sign in, which my account cannot do. I use Python 3 on macOS directly for the scripts.